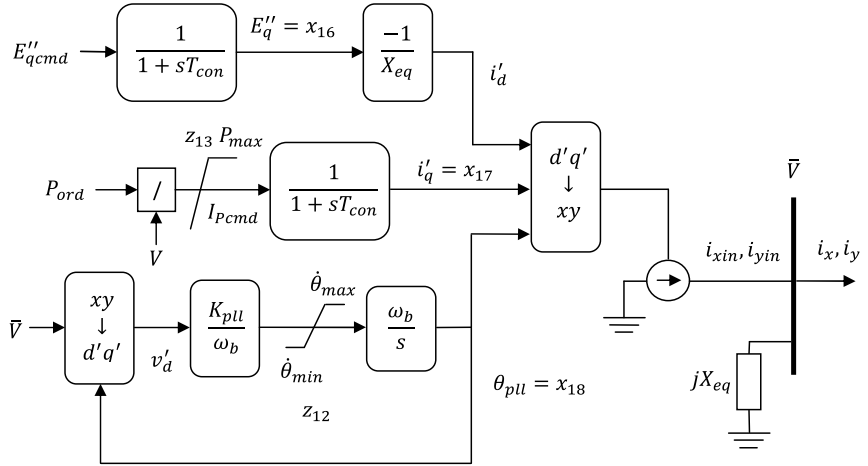


WT3 (DFIG) Modelling in RAMSES



Network and injector currents relationship in DFIG base:

$$i_{xin} = i'_q \cos(\theta_{pll}) + \frac{E''_q}{X_{eq}} \sin(\theta_{pll})$$

$$i_{yin} = i'_q \sin(\theta_{pll}) - \frac{E''_q}{X_{eq}} \cos(\theta_{pll})$$

$$0 = -i_{yin} + i_y \frac{S_{base}}{S_{nom}} - \frac{v_x}{X_{eq}}$$

$$0 = -i_{xin} + i_x \frac{S_{base}}{S_{nom}} + \frac{v_y}{X_{eq}}$$

$$v_d = -v_x \sin(\theta_{pll}) + v_y \cos(\theta_{pll})$$

with $E''_q = x_{16}$, $i'_q = x_{17}$, and $\theta_{pll} = x_{18}$